

STAT 20000: Elementary Statistics

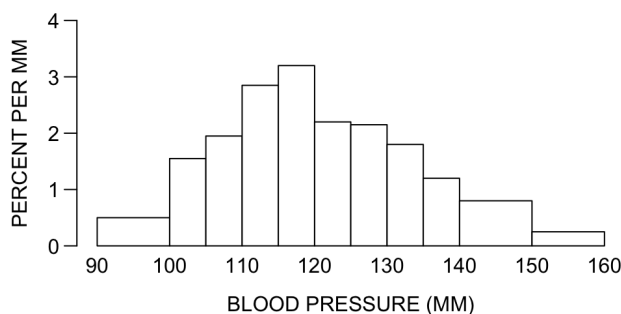
Homework 2

Due: 11:59 pm, April 10, 2026

All page, section, and exercise numbers below refer to the course text *Statistics*, 4th edition, by D.Freedman, R. Pisani & R. Purves (FPP).

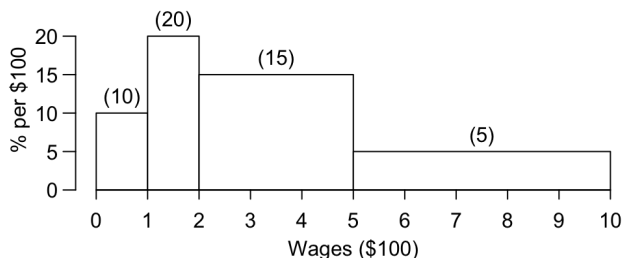
Reading: Chapter 3-4 in FPP.

Q1 (5 points). (Chapter 3, Review Exercise 4 on p.51-52, Revised) The figure below is a histogram showing the distribution of blood pressure for all 14148 women in the Drug Study (section 5). Use the histogram to answer the following questions.

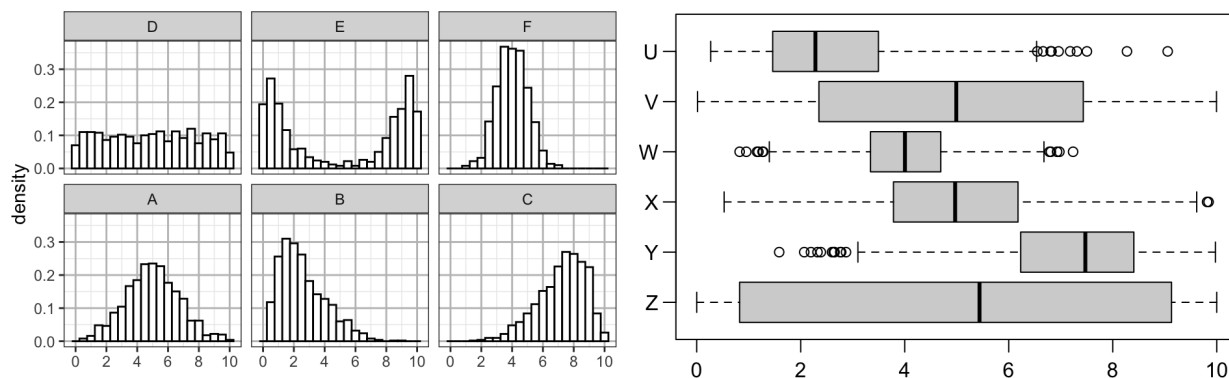


- On the interval 90-100 mm, the height of the histogram is about 0.5% per mm. What percentage of the women had blood pressures in this class interval?
- In which interval are there more women: 135-140 mm or 140-150 mm?
- Which interval is more crowded (have a higher density): 135-140 mm or 140-150 mm?
- Which interval has more women: 97-98 mm or 102-103 mm?
- Is the percentage of women with blood pressures below 110 mm closest 20%, 40%, 60%, or 80%?

Q2 (2 points) (Exercise 1 in Exercise Set C on p.41, Revised) A histogram of monthly wages for part-time employees is shown below (densities are marked in parentheses). Find the **median** monthly wages for part-time employees.



Q3 (6 points). Below are the histograms and boxplots for 6 different data sets, in scrambled order. Match each of the histograms (A, B, C, D, E, F) with its corresponding boxplot (U, V, W, X, Y, Z).



Q4 (6 points) (Chapter 4, Review Exercise 6, on p.75) Below are sketches of histograms for three lists.

- (2 points) In scrambled order, the averages are 40, 50, 60. Match the histograms with the averages.
- (2 points) Match the histogram with the description:
 - the median is less than the average
 - the median is about equal to the average
 - the median is bigger than the average
- (2 points) Is the SD of histogram (iii) around 5, 15, or 50?

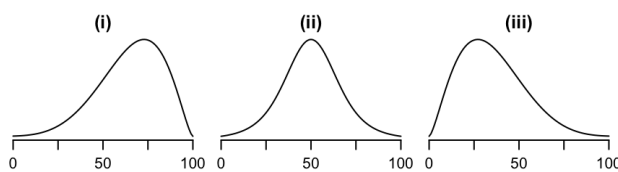


Figure 1: Caption

Q5 (6 points) The average monthly salary of all employees in a certain company is \$6,000, with an SD of \$3,000.

- As the revenue of the company increased 50% this year, the executive proposes to give all employees a \$3000 bonus next month. What is the average amount of money that employees in this company can receive next month (regular monthly salary + bonus), and the SD?
- Another proposal is to give each employee a bonus equal to 50% of his/her regular monthly salary. Based on this proposal, what is the average amount of money that employees in this company can receive next month (regular monthly salary + bonus), and what is the the SD?

- (c) The three sketches below show the change of the histograms for the amount of money of that employees can receive this month (regular monthly salary only) and the next month (regular monthly salary + bonus). One corresponds to a flat \$3000 bonus for everyone, one to a bonus equal to 50% of the monthly salary, and one is wrong. Which one is which? Explain briefly.

